CHAPTER 4

4.1) NUMBER OF THERMAL PHOTONS

We need

$$\sum_n \langle s_n \rangle = \sum_n 1/[\exp(\hbar\omega_n/\tau) - 1] = \sum_n 1/(e^u - 1) \quad ,$$

where $\omega_n = n\pi c/L$, $u = (\pi\hbar c/\tau L)n$. The sum is converted to an integral, as in (17), but with an additional factor 2 for the two independent polarizations:

$$\sum_n \ldots = \pi \int_0^\infty \ldots\ n^2 dn = \pi(\tau L/\pi\hbar c)^3 \int_0^\infty \ldots\ u^2 du \quad .$$

The factor preceding the final integral may be written $\pi^{-2}V(\tau/\hbar c)^3$, as in (48). The integral has the value, evaluated below,

$$\int_0^\infty \frac{u^2 du}{e^u - 1} = 2.404113806 \quad .$$

<u>Comment</u>. Integrals of the form

$$I = \int_0^\infty \frac{u^n du}{e^u - 1} \quad \text{with } n \geqq 0$$

occur often in thermal physics. They may be evaluated by expanding the denominator into a (converging) geometric series:

$$I_n = \sum_{m=1}^\infty \int_0^\infty e^{-mu} u^n du = \sum_{m=1}^\infty \frac{1}{m^{n+1}} \int_0^\infty e^{-v} v^n dv = n! \sum_{m=1}^\infty \frac{1}{m^{n+1}} \quad .$$

For odd integer values of n the sums have simple closed-form values ($I_1 = \pi^2/6$, $I_3 = \pi^2/15$), listed in many tables. For other values no such simple relations exist. Purely numerical values for integer n are given in many of the more elaborate tables, but the series are readily (although slowly) summed on a programmable calculator. The calculation can be greatly speeded up by summing only a relatively small number of terms and approximating the remainder as an integral:

$$\sum_{m=1}^{\infty} \frac{1}{m^{n+1}} \cong \sum_{m=1}^{M} \frac{1}{m^{n+1}} + \int_{M+\frac{1}{2}}^{\infty} \frac{dm}{m^{n+1}} \quad .$$

The integral has the value $1/[n(M+\frac{1}{2})^n]$. For $n = 2$, $M = 10$ one obtains in this way $I_2 = 2.404134$, a result within less than 1 part in 10^5 of the exact value, 2.404114. The accuracy increases with increasing M.

4.2) <u>SURFACE TEMPERATURE OF THE SUN</u>

(a) Let J_E denote the radiated energy flux density at the Sun-to-Earth distance D_E from the Sun. Then the total radiated flux is

$$\Phi = 4\pi D_E^2 J_E = 3.845\times10^{26} \text{J s}^{-1} \quad ,$$

(b) If $R_\odot$ is the radius of the Sun and $J_\odot$ the flux density at that radius,

$$J_\odot = \Phi/4\pi R_\odot^2 = \sigma_B T^4 = 6245 \text{ J s}^{-1}\text{cm}^{-2} \quad ,$$

$$T = (J_\odot/\sigma_B)^{\frac{1}{4}} = 5761 \text{ K} \quad .$$

4.3) <u>AVERAGE TEMPERATURE OF THE INTERIOR OF THE SUN</u>

(a) Dimensionally,

$$U \cong - GM_\odot^2/R = - 3.77\times10^{48}\text{erg} \cong - 4\times10^{48}\text{erg} \quad . \qquad \text{(S1)}$$

If the Sun had a uniform density $\rho = M_\odot/(4\pi R_\odot^3/3)$, the exact result would be

$$U = -G\rho^2 \int_0^\infty (4\pi R^3/3)(4\pi R^2)dR = -3GM_\odot^2/5R_\odot \quad .$$

We continue with the value (S1).

(b) The Sun consists mostly of hydrogen atoms; their number is

$$N \cong M_\odot/M_H = 1.195\times10^{57} \cong 1\times10^{57} \quad .$$

From $3\tau N/2 = -U/2$:

$$T = -U/3k_BN = 9.66\times10^6 K \cong 10^7 K \quad .$$

4.4) <u>AGE OF THE SUN</u>

(a) The reaction stops when the number of ^4He atoms reaches $N = 0.1\times M_\odot/M_{He} \cong 3.0\times10^{55}$. Next,

$$\Delta M = (4\times1.0078 - 4.0026) \times 1 \text{ amu} \cong = 0.09818 \text{ amu}$$

$$\cong 1.63\times10^{-25} g. \quad \text{Then,}$$

$$E = N \times \Delta M \times c^2 \cong 1.28\times10^{44} J \quad .$$

(b) Let Φ denote the total radiated energy flux. Then

$$t = E/\Phi \cong 3.2\times10^{17} s \cong 1.0\times10^{10} yr \quad .$$

(useful mnemonic device: 1 yr $\cong \pi\times10^7$s, to within better than ½%. The average year, on the Gregorian calendar, has 365.2425 days = 3.1557×10^7s).

4.5) <u>SURFACE TEMPERATURE OF THE EARTH</u>

The solar energy flux density at the Earth's orbit is $J_E = \sigma_B T_\odot^4 \times (R_\odot/D_E)^2$. The flux intercepted by the earth is

$$\Phi = \pi R_E^2 J_E = \pi\sigma_B T_\odot^4 \times (R_E R_\odot/D_E)^2 \quad .$$

The flux re-radiated by the Earth is

$$\Phi = \sigma_B T_E^4 \times 4\pi R_E^2 \quad .$$

Equating the two fluxes leads to

$$T_E = (R_\odot/2D_E)^{\frac{1}{2}} T_\odot \cong 280 \text{ K} \quad .$$

Note that the radius of the Earth drops out.

Comment. The problem lends itself to various elaborations, such as estimating the temperature of the solar energy panels of a satellite, as a function of the attitude of the panels, and of the emissivity of the back of the panel.

4.6) PRESSURE OF THERMAL RADIATION

(a) $U = \sum_j s_j \hbar\omega_j$. If the system is kept in the same state during a volume change, the occupation numbers remain the same and the change is isentropic. With $(\partial s_j/\partial V)_\sigma = 0$:

$$p = -(\partial U/\partial V)_\sigma = -\sum_j s_j \hbar (d\omega_j/dV) \quad . \tag{50}$$

We have written $d\omega_j/dV$ rather than $(\partial\omega_j/\partial V)_\sigma$, because the mode frequencies depend only on the cavity dimensions.

(b) We assume an isotropic volume change of a cube-shaped cavity (see comment below). From (15):

$$\log \omega_j = -\log L + \text{constant} = -\frac{1}{3}\log V + \text{constant},$$

$$\frac{d\omega_j}{\omega_j} = -\frac{1}{3}\frac{dV}{V} \quad \text{or} \quad \frac{d\omega_j}{dV} = -\frac{1}{3}\frac{\omega_j}{V} \quad . \tag{51}$$

(c) Insert (51) into (50):

$$p = +\frac{1}{3V}\sum_j s_j \hbar\omega_j = U/3V \quad . \tag{52}$$

(d) Insert (20) into (52):

$$p = \frac{\pi^2 k_B^4 T^4}{45\hbar^3 c^3} = \frac{4\sigma_B}{3c} T^4 \quad .$$

This is equal to the gas pressure, $p = nk_BT$, when

$$T = (3cnk_B/4\sigma_B)^{1/3} \quad .$$

For $n = N_A/1\mathrm{cm}^3 \cong 6.0\times10^{23}\mathrm{cm}^{-3}$: $T \cong 3.2\times10^7\mathrm{K}$.

Comment. The result (51), while not depending on the exact shape of the cavity, requires that the volume change of the cavity is isotropic. For an anisotropic volume change the relative change in frequency is different for different modes but the result (52) remains valid anyway. Students interested in rigor might be encouraged to work out the details, but the subject is ill-suited for general classroom use.

4.7) FREE ENERGY OF A PHOTON GAS

(a) The partition function for a single mode with frequency ω_n is, from (3),

$$Z_n = [1 - \exp(-\hbar\omega_n/\tau)]^{-1} \quad . \tag{S1}$$

The different modes are independent of each other; therefore the partition function for the overall photon gas is simply the product of all single-mode partition functions:

$$Z = \prod_n Z_n \quad .$$

Because of (S1) this is the same as (53).

(b) $F = -\tau \log Z = -\tau \quad \log Z_n$. With (S1) this is the same as (54). Transformation to an integral:

$$\sum_n \cdots = \pi \int_0^\infty \cdots \; n^2 dn = \pi(\tau L/\pi\hbar c)^3 \int_0^\infty \cdots \; u^2 du \quad ,$$

with $u = (\pi\hbar c/\tau L)n$; see Problem 4.1. Integration by parts:

$$\int_0^\infty \log(1-e^{-u})u^2du = \frac{u^3}{3}\log(1-e^{-u})\Big|_0^\infty - \frac{1}{3}\int_0^\infty \frac{u^3du}{e^u-1} = -\frac{\pi^4}{45} \quad .$$

The integrated term vanishes at both $u = 0$ and $u = \infty$, and the integral is the same as in (19), $\pi^4/15$. Inserted into F:

$$V = \tau \times \pi(\tau L/\pi\hbar c)^3 \times (-\pi^4/45) = -\pi^2 V\tau^4/45c^3\hbar^3 \quad .$$

4.8) HEAT SHIELDS

We treat the general case of N heat shields ($n = 1, \ldots, N$), in the order $T_\ell \equiv T_0 < T_1 < \ldots < T_N < T_{N+1} \equiv T_u$. The radiative energy flux density between planes n and n-1 is $J_U = \sigma_B(T_n^4 - T_{n-1}^4)$, independent of n. Hence

$$T_n^4 = T_{n-1}^4 + J_U/\sigma_B = T_{n-2}^4 + 2J_U/\sigma_B = T_{n-m}^4 + mJ_U/\sigma_B$$

$$= T_\ell^4 + nJ_{U/\sigma B} \quad . \qquad (S1)$$

Applied to $n = N+1$ and solved for J_U:

$$J_U = \sigma_B(T_u^4 - T_\ell^4)/(N+1) \quad .$$

Inserted into (S1):

$$T_n^4 = T_\ell^4 + \frac{n}{N+1}(T_u^4 - T_\ell^4) = [(N+1-n)T_\ell^4 + nT_u^4]/(N+1) \quad .$$

For $n = N = 1$ we have $J_U = \sigma_B(T_u^4 - T_\ell^4)/2$, $T_1^4 = (T_\ell^4 + T_u^4)/2$.

Comment. The instructor might suggest the more general case for extra credit.

4.9) PHOTON GAS IN ONE DIMENSION

$$U = \sum_n \langle s(\omega_n)\rangle \hbar\omega_n = \sum_n \hbar\omega/[\exp(\hbar\omega_n/\tau)-1]$$

$$\cong \tau \int_0^\infty \frac{(\hbar\omega_n/\tau)dn}{\exp(\hbar\omega_n/\tau)-1} \quad .$$

The mode frequencies are those for which the length of the line is an integer multiple of $\lambda/2$: $\lambda = 2L/n$, $\omega_n = 2\pi v/\lambda = (\pi v/L)n$. Let $u = \hbar\omega_n/\tau = (\pi\hbar v/\tau L)n$. Then,

$$U = (\tau^2 L/\pi\hbar v)\int_0^\infty \frac{udu}{e^u-1} \quad .$$

The integral has the value $\pi^2/6$. (See, for example, Dwight, #860.31). Therefore

$$U = \pi L\tau^2/6\hbar v \quad ,$$

$$C_L = (\partial U/\partial\tau)_L = \pi L\tau/3\hbar v \quad .$$

Comment. The integral can be evaluated numerically without reference to a table, by the method discussed in the solution to Problem 4.1. For M = 10 one obtains 1.6450, which agrees with the exact value (1.64493) to within about 4 parts in 10^5.

4.10) HEAT CAPACITY OF INTERGALACTIC SPACE

For the atoms, $C_H/V = (3/2)nk_B$. For photons, we first re-express the proportionality factor in the expression (20) by inserting the Stefan-Boltzmann constant, $\sigma_B = \pi^2 k_B^4/60\hbar^3 c^2$:

$$U/V = 4\sigma_B T^4/c \quad , \quad C_{rad}/V = 16\sigma_B T^3/c \quad ;$$

$$C_H/C_{rad} = \frac{3ck_B}{32\sigma_B}\frac{N/V}{T^3} \cong 2.84\times10^{-10} \quad .$$

Comment. It usually simplifies numerical calculations if one expresses U/V in the form used here, involving σ_B/c and T, rather than the form (20) directly.

4.11) HEAT CAPACITY OF SOLIDS IN HIGH-TEMPERATURE LIMIT

With the help of (44),(41) may be written as

$$U = 9nk_B\theta \times \frac{I}{x_D^4} \int_0^{x_D} \frac{x^3 dx}{e^x - 1} \ , \text{ with } x_D = \theta/T \quad .$$

When $T \gg \theta$, $x_D \ll 1$, and we can expand for all $x \leq x_D$:

$$e^x - 1 = x + x^2/2! + x^3/3! \ \ldots = x(1 + y) \quad ,$$

where $y = x/2 + x^2/6 + \ldots \ll 1$. Then

$$\frac{x^3}{e^x - 1} = \frac{x^2}{1 + y} = x^2(1 - y + y^2 - y^3 + \ldots) \quad .$$

If we keep only terms up to x^2,

$$y \cong x/2 + x^2/6 \quad , \ y^2 \cong x^2/4 \quad , \ y^3 \cong 0 \quad .$$

With these:

$$x^3/(e^x - 1) \cong x^2(1 - x/2 + x^2/12) \quad ,$$

$$\frac{1}{x_D^4} \int_0^{x_D} \frac{x^3 dx}{e^x - 1} \cong \frac{1}{x_D^4} (x_D^3/3 - x_D^4/8 + x_D^5/60) = \frac{T}{3\theta} - \frac{1}{8} + \frac{\theta}{60T} \quad ,$$

$$U = 3Nk_B[T - 3/8\theta + \theta^2/20T] \quad ,$$

$$C_V = 3Nk_B[1 - \theta^2/20T^2] \quad .$$

For $T = 1.4\theta$ we have $\theta^2/20T^2 \cong 0.0255$; that is, C_V is about 2.5% below its asymptotic value, in rough agreement with Fig. 4.11. Fig. 4.11 does not extend to $T = 1.4\theta$; use Table 4.2 for $\theta/T = 0.7 \cong 1/1.43$ instead, yielding $1 - 24.34/24.943 \cong 0.024$.

4.12) HEAT CAPACITY OF PHOTONS AND PHONONS

For phonons, for $T << \theta$ from, (47b):

$$C_V/V = (12\pi^4/5)\ nk_B\ (T/\theta)^3 \cong 322\ \text{erg}\ K^{-1}\text{cm}^{-3} \quad .$$

For photons, as in Problem 4.10,

$$U/V = 4\sigma_B T^4/c \quad ,$$
$$C_V/V = 16\sigma_B T^3/c = 2.99\times10^{-14}\text{erg}\ K^{-4}\text{cm}^{-3} \times T^3 \quad .$$

The two values are equal for

$$T = (322/2.99\times10^{-14})^{1/3}\ K \cong 2.2\times10^5 K \quad .$$

4.13) ENERGY FLUCTUATIONS IN A SOLID AT LOW TEMPERATURES

With the help of (3.89),

$$\mathscr{F}^2 \equiv \langle(\Delta U)^2\rangle/U^2 = \tau^2(\partial U/\partial\tau)_V/U^2 \quad .$$

In the low-temperature range $U \propto \tau^4$, hence $(\partial U/\partial\tau)_V$ $4U/\tau$, $\mathscr{F}^2 = 4\tau/U$. Insert (46) for U:

$$\mathscr{F}^2 = \frac{20}{3\pi^4}\,\frac{1}{N}\left(\frac{k_B\theta}{\tau}\right)^3 = 0.684 \times \frac{1}{N}\left(\frac{\theta}{T}\right)^3 \quad ,$$

$$\mathscr{F} = 0.262 \times \left[\frac{1}{N}\left(\frac{\theta}{T}\right)^3\right]^{\frac{1}{2}} \quad .$$

4.14) HEAT CAPACITY OF LIQUID ^{4}He AT LOW TEMPERATURES

(a) We need N/V:

$$N/V = \frac{6.02\times10^{23}\text{mol}^{-1} \times 0.145\ \text{g/cm}^3}{4\ \text{g/mol}} = 2.18\times10^{27}\text{atoms/cm}^3 \quad .$$

With this, and $v = 2.383\times10^4\text{cm s}^{-1}$, from (44),

$$\theta = (\hbar v/k_B)(6\pi^2 N/V)^{1/3} = 19.83K \quad .$$

(b) For 1 gram, $N = 6.02\times10^{23}/4 \cong 1.5\times10^{23}$. With this and the above value for θ, from (47b), after dividing by 3, to account for the absence of transverse modes,

$$C_V(\text{per gram}) = \frac{1}{3}\,\frac{12\pi^4}{5}\,Nk_B(T/\theta)^3$$

$$= 0.0208\ \text{J g}^{-1}\text{K}^{-4} \times T^3 \quad ,$$

which is very close to the experimental value.

Comment. Note that we accounted for the absence of the longitudinal mode by dividing the final heat capacity by 3, not by adjusting the Debye temperature to the lower number of contributing modes.

4.15) ANGULAR DISTRIBUTION OF RADIANT ENERGY FLUX

(a) The thermal radiation within the black body cavity itself is isotropic. All energy travels with the speed c, but the directions of travel are uniformly distributed over a solid angle 4π, hence the energy flux density whose direction of travel falls within a solid angle $d\Omega$ is $(cU/V) \times d\Omega/4\pi$.

Suppose now that the direction of travel makes an angle θ with the direction normal to the plane of a small hole of area A in the wall of the cavity. This hole will let pass the same amount of radiation as a hole of the size $A \cos\theta$ perpendicular to the direction of radiation travel. Hence the energy flux emerging through the hole into the solid angle $d\Omega$ is $cU/V \cos\theta\ d\Omega/4\pi$.

(b) The total energy flux emerging through the hole is obtained by integrating over all forward angles, with $d\Omega = \sin\theta\ d\theta\ d\phi$:

$$\Phi = \frac{cUA}{4\pi V}\int \cos\theta\ d\Omega = \frac{cUA}{4\pi V}\int_0^{2\pi} d\phi \int_0^{\pi/2} \cos\theta\ \sin\theta\ d\theta \quad .$$

The integration over ϕ yields 2π. The integral over θ has the value 1/2. Hence

$$J_U = \Phi/A = \frac{c}{4}\frac{U}{V} \quad .$$

We have given here the argument for the total radiation, integrated over all frequencies. The argument applies to every frequency interval $(\omega,\omega+d\omega)$ individually .

4.16) <u>IMAGE OF A RADIANT OBJECT</u>

Assume that the solid angles Ω_H and Ω_0 are centered about the directions perpendicular to the areas A_H and A_0, and that they are sufficiently small, so we may neglect the cosine falloff of the flux density with increasing angle from the normal direction discussed in Problem 4.15. The total energy flux intercepted by the lens and directed onto the object is then $J_U(\tau_H)A_H\Omega_H/4\pi$. Suppose now that the object is itself a hole in another black body, with area A_0. This second black body then radiates the energy flux $J_U(\tau_0)A_0\Omega_0/4\pi$ toward the first body. If the two bodies are at the same temperature, $\tau_0 = \tau_H$, they are in thermal equilibrium with each other, and the two fluxes must cancel, just as in the Kirchhoff law argument in the text. Thus $A_H\Omega_H = A_0\Omega_0$.

This relation implies, amongst other things, that it is not possible to heat any object by radiant heat to a temperature higher than that of the heat source, even if the heat source is focused upon the object.

For large solid angles, for oblique optical systems, and in the presence of diffraction, the relation relating sizes and solid angles, for objects and their images, becomes more complicated, but an analogous relation can always be derived.

4.17) ENTROPY AND OCCUPANCY

We draw on (3.49) and (3.55) to express σ in terms of Z:

$$\sigma = - \partial F/\partial\tau = + \partial(\tau \log Z)/\partial\tau$$
$$= \log Z + (\tau/Z)(\partial Z/\partial\tau) \quad . \tag{S1}$$

Here Z is given by (3),

$$Z = 1/[1 - \exp(-\hbar\omega/\tau)] \quad . \tag{3}$$

We re-express this in terms of $\langle s\rangle$, by utilizing Eq. (6):

$$\langle s\rangle = 1/[\exp(\hbar\omega/\tau) - 1] \quad . \tag{6}$$

From (3) and (6) follow:

$$\langle s+1\rangle = \langle s\rangle + 1 = \exp(\hbar\omega/\tau)/[\exp(\hbar\omega/\tau) - 1]$$
$$= 1/[1 - \exp(-\hbar\omega/\tau)] = Z \quad , \tag{S2}$$

$$\langle s+1\rangle/\langle s\rangle = \exp(\hbar\omega/\tau) \quad ,$$
$$\hbar\omega/\tau = \log[\langle s+1\rangle/\langle s\rangle] \quad , \tag{S3}$$

$$\frac{\partial Z}{\partial\tau} = \frac{\partial\langle s+1\rangle}{\partial\tau} = \frac{\partial\langle s\rangle}{\partial\tau} = - \frac{\exp(\hbar\omega/\tau)}{[\exp(\hbar\omega/\tau)-1]^2} \times \frac{-\hbar\omega}{\tau^2}$$

$$= \frac{1}{\tau}\langle s+1\rangle\langle s\rangle\log[\langle s+1\rangle/\langle s\rangle] \quad . \tag{S4}$$

We insert (S2) and (S4) into (S1):

$$\sigma = \log\langle s+1\rangle + \frac{\tau}{\langle s+1\rangle} \times \frac{1}{\tau} \langle s+1\rangle\langle s\rangle \log[\langle s+1\rangle/\langle s\rangle]$$
$$= \log\langle s+1\rangle + \langle s\rangle[\log\langle s+1\rangle - \log\langle s\rangle]$$
$$= \langle s+1\rangle\log\langle s+1\rangle - \langle s\rangle\log\langle s\rangle \quad . \tag{58}$$

4.18) ISENTROPIC EXPANSION OF PHOTON GAS

(a) With the volume being proportional to R^3, the isentropic relation $\tau V^{1/3}$ = constant implies τR = constant, or

$$R_i/R_f = \tau_f/\tau_i \cong 2.9\text{K}/3000\text{K} \cong 10^{-3} \quad .$$

The final temperature $T_f = 2.9K$ is the present cosmic black-body temperature (see Figure 4.5).

(b) $W = U_i - U_f$. From (20), $U = s_B V\tau^4$, where $s_B = \pi^2/15\hbar^3c^3$. Hence $W = s_B(V_i\tau_i^4 - V_f\tau_f^4)$. Now, $V_i\tau_i^3 = V_f\tau_f^3$. Therefore $V_f\tau_f^4 = V_i\tau_i^3\tau_f$ and

$$W = s_B(V_i\tau_i^4 - V_i\tau_i^3\tau_f) = s_B V_i\tau_i^3(\tau_i - \tau_f) \quad .$$

4.19) <u>REFLECTIVE HEAT SHIELD AND KIRCHHOFF'S LAW</u>

Let τ_i be the (unknown) temperature of the intermediate sheet. The net flux density from τ_u to τ_i is

$$J_u = \sigma_B\tau_u^4 - e\sigma_B\tau_i^4 - r\sigma_B\tau_u^4 = a\sigma_B(\tau_u^4 - \tau_i^4) \quad ,$$

because $a = e$ and $r = (1 - a)$. The net flux density from τ_i to τ_ℓ is

$$J_\ell = e\sigma_B\tau_i^4 - \sigma_B\tau_\ell^4 + r\sigma_B\tau_\ell^4 = a\sigma_B(\tau_i^4 - \tau_\ell^4) \quad .$$

In steady state $J_u = J_\ell$, whence the temperature τ_i satisfies

$$2a\sigma_B\tau_i^4 = a\sigma_B(\tau_u^4 + \tau_\ell^4) \text{ or } \tau_i^4 = \tfrac{1}{2}(\tau_u^4 + \tau_\ell^4) \quad ,$$

the same as for a black heat shield. The net flux is obtained by inserting this value of τ_i into the expression for J_u or J_ℓ:

$$J = \tfrac{1}{2}(1 - r)\sigma_B(\tau_u^4 - \tau_\ell^4) \quad .$$